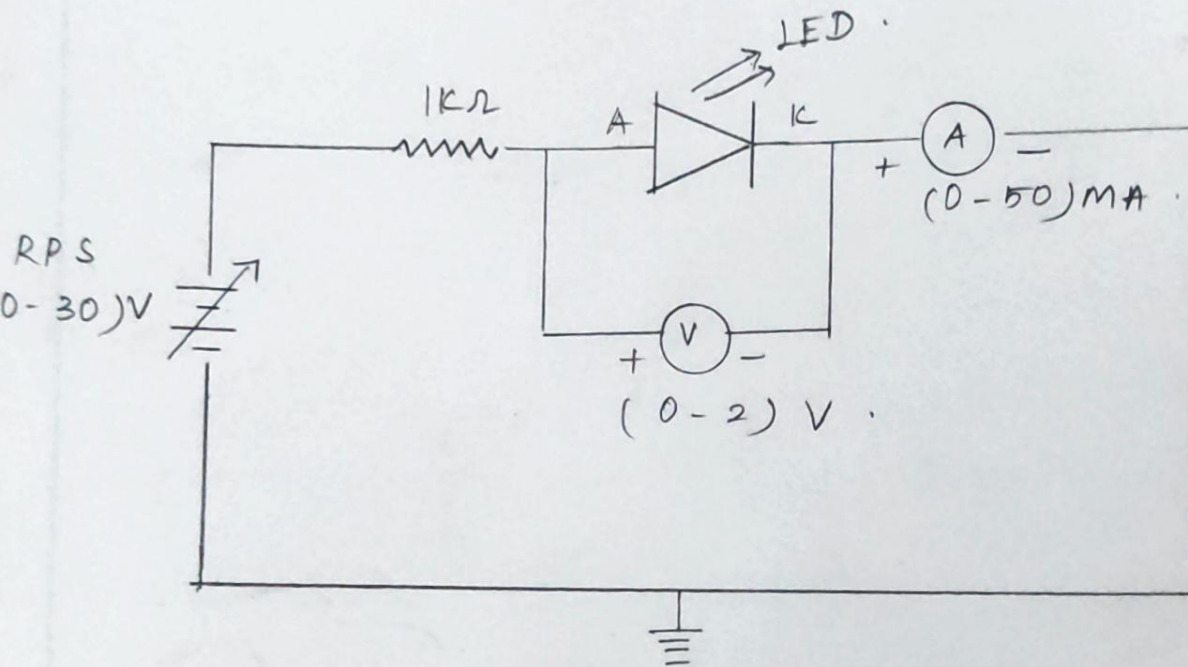


CIRCUIT DIAGRAM :-

LIGHT EMITTING DIODE

FORWARD BIAS.



EX NO.: 4

CHARACTERISTICS OF PHOTONIC DEVICES .

25.7.17

a) LED

Aim :

To determine the determination of V-I characteristics of LED .

Apparatus Required:

Sl No.	Apparatus Name	Type / Range	Quantity
1.	Regulated power supply (RPS)	(0-30) V	1
2.	Voltmeter V_1	(0-20) V	1
3.	Voltmeter V_2	(0-2) V	1
4.	Voltmeter V_3	(0-10) V	1
5.	Ammeter I_1	(0-50) mA	1
6.	Ammeter I_2		1
7.	Bread Board .	-	1

Component Required:

Sl No.	Apparatus Name	Type / Range	Quantity .
8.	Light Emitting Diode	-	1
9.	Electric bulb with holder .	100 W	1
10.	connecting wires	-	As required .



TABULATION :

LIGHT EMITTING DIODE

GREEN

RED

YELLOW

$V_F(V)$	$I_F(mA)$	$V_F(V)$	$I_F(mA)$	$V_F(V)$	$I_F(mA)$
0.30	0.0	0.34	0.00	0.23	0.00
0.76	0.0	0.69	0.00	0.63	0.00
1.02	0.0	1.00	0.00	1.02	0.00
1.18	0.0	1.25	0.00	1.26	0.00
1.38	0.0	1.66	0.00	1.44	0.00
1.67	0.0	1.89	0.06	1.79	0.00
1.90	0.55	1.97	0.25	1.94	0.18
1.94	0.91	2.01	0.53	2.00	0.46
1.97	1.10	2.06	0.96	2.03	0.80
1.98	1.46	2.08	1.27	2.05	1.20
2.01	1.91	2.09	1.34	2.06	1.34
2.02	1.94	2.12	1.69	2.08	1.64
2.02	2.20	2.12	1.99	2.09	1.94
		2.12	2.24	2.09	2.2
				2.11	2.58
				2.13	2.72
				2.14	3.09
				2.15	3.47
				2.15	3.66
				2.16	4.07

I_P (mA)

LIGHT EMITTING DIODE.

SCALE

X AXIS 1 cm = 0.2 V

Y AXIS 1 cm = 0.4 (mA)



0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4 2.6

MODEL GRAPH:-

I_F (mA)



RED

YELLOW

GREEN

V_F (V)



Procedure:

- * Connection is made as per the diagram.
- * The distance between the light source and phototransistor is fixed at a constant level and by varying input voltage corresponding emitter current values are noted.
- * This values procedure is repeated for different values of distances from the light source.
- * All the readings are tabulated.
- * A graph is plotted between V_{CE} and I_E for different values of length (L).

RESULT:

Experiment

Thus the connection for LED is given and characteristics was obtained and graphs are plotted below.

